PHYSICS MOST REPEATED

Targeted Question List Based on Current Board Patterns (2023-2025 Sets)

1. SEMICONDUCTORS & DEVICES ~7 Marks

The Rectifier Question:

Explain the formation of the **depletion region** and **potential barrier**. Draw the circuit diagram of a **Full Wave Rectifier** and explain its working with input/output waveforms.

> Variation: V-I characteristic graphs for forward and reverse bias often appear in Section B.

2. OPTICS (RAY & WAVE) ~15 Marks

Optical Instruments:

Draw a labelled ray diagram for a **Compound Microscope** or **Astronomical Telescope** (Normal Adjustment). Derive the expression for its **magnifying power**.

Wave Nature (Huygens):

Define wavefront. Use **Huygens' Principle** to verify either Snell's Law of **Refraction** or the Law of **Reflection**.

> Numerical Focus: Calculate fringe width change in YDSE when the apparatus is immersed in water ($ eta' = eta / \mu $).

3. MODERN PHYSICS (ATOMS & DUAL NATURE) ~10 Marks

Bohr Model:

Obtain the expression for the **radius of the n-th orbit** ($r \propto n^2$) or the total **energy of an electron** using Bohr's postulates.

Photoelectric Effect:

Analyze a graph of **stopping potential (V_0) vs. frequency (v)**. Determine threshold frequency and Planck's constant from slope/intercept.

4. MAGNETISM & CURRENT

~15 Marks

Galvanometer:

Principle and working of a **Moving Coil Galvanometer** (MCG). Explain the role of the **radial magnetic field** and convert MCG to a Voltmeter/Ammeter.

Drift Velocity:

Derive the relation between **current (I) and drift velocity (v_d)** ($I = n e A v_d$). Explain the temperature effect on resistivity.

SUMMARY: TOP 6 "MUST-STUDY" DERIVATIONS

- ✔ **Full Wave Rectifier** (Circuit + Output)
- ✔ **Telescope/Microscope** Ray Diagrams
- ✔ **Snell's Law** via Huygens Wavefronts
- ✔ **Bohr's Orbit Radius** (r_n)
- ✔ **Current-Drift Velocity** Relation ($I = n e A v_d$)
- ✔ **Electric Dipole Torque** ($ au = pE \sin heta$)